

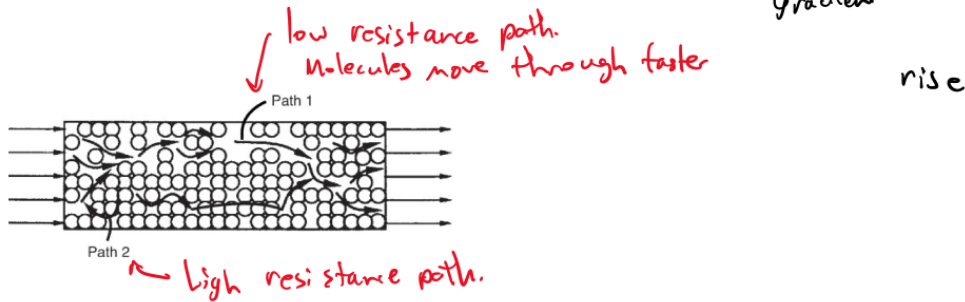
29 Mixing in reactors and residence time distributions

So far in this class, we have studied ideal reactors, i.e., where we have assumed perfect mixing

For example:

- In the batch reactor and CSTR, we have assumed the contents of the entire tank have the exact same concentration, temperature, etc, i.e., that there are no gradients over the entire volume
- In the PFR and PBR, we have assumed perfect mixing in the radial dimension & no mixing in the axial dimension, i.e.  no mixing between adjacent volume slices
entire volume has the exact same concentration, temp, etc, no gradients

Real reactors are generally not perfectly mixed. For example:

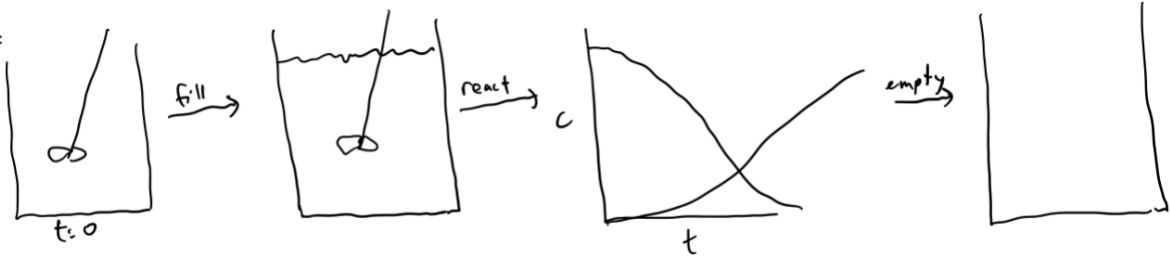


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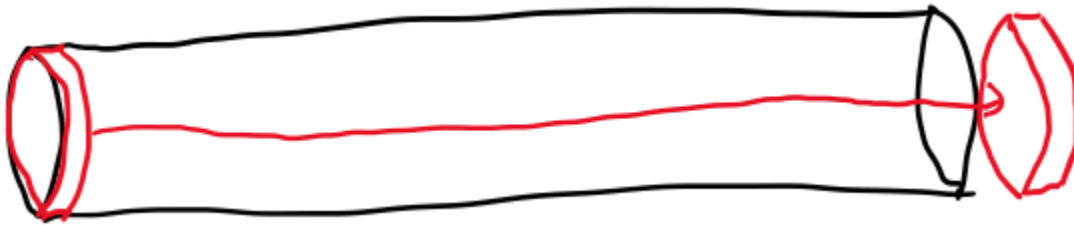
In these examples, different molecules spend *different* amounts of time in the reactor

In ideal PFRs, PBRs, and batch reactors, all the molecules spend *the same* amount of time in the reactor

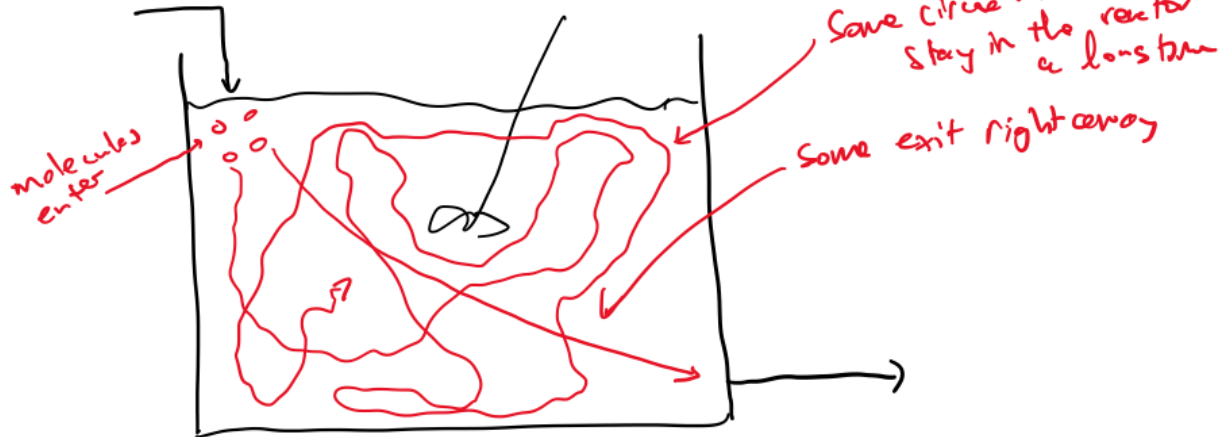
- Batch:



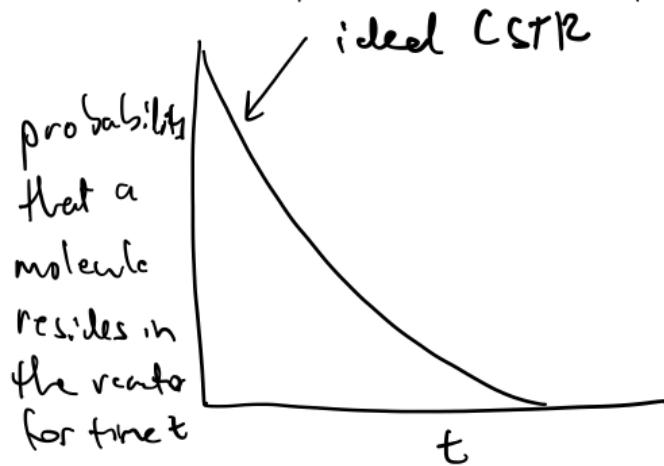
- PFR/PBR:



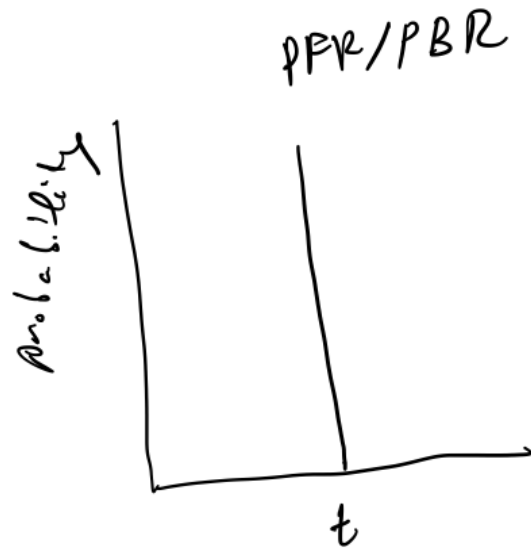
In contrast, there is a distribution of residence times in a CSTR:



Gives rise to the concept of residence time distribution (RTD):



Residence time distributions also occur in non-ideal PFRs/PBRs.



Conversions in real reactors are usually lower than in ideal reactors.

Lower conversions are caused by ineffective contacting btwn molecules